

Solutions to UGB Mock 3

Q. 1

Problem: Prove that, for any natural number n , there exists a n digit number which contains only the digits 2 and 3, and is divisible by 2^n .

Proof: We will prove this by induction on n .

Base Case: For $n = 1$, the number 2 is a 1-digit number consisting only of the allowed digits, and it is perfectly divisible by 2^1 .

Inductive Step: Assume that for some natural number n , there exists an n -digit number A_n composed only of the digits 2 and 3 such that A_n is divisible by 2^n . This implies we can write $A_n = k \cdot 2^n$ for some integer k .

We can form two new $(n + 1)$ -digit numbers by prepending either a 2 or a 3 to A_n :

1. $B = 2 \cdot 10^n + A_n$

2. $C = 3 \cdot 10^n + A_n$

Factoring out 2^n from both expressions gives:

$$B = 2^{n+1} \cdot 5^n + k \cdot 2^n = 2^n(2 \cdot 5^n + k)$$

And,

$$C = 3 \cdot 2^n \cdot 5^n + k \cdot 2^n = 2^n(3 \cdot 5^n + k)$$

Since 5 is an odd number, any power 5^n is also odd. Thus, $2 \cdot 5^n$ is an even number, and $3 \cdot 5^n$ is an odd number. Because $(2 \cdot 5^n)$ and $(3 \cdot 5^n)$ have different parities, the sum $(2 \cdot 5^n + k)$ and $(3 \cdot 5^n + k)$ must also have different parities, regardless of whether k is even or odd.

Therefore, exactly one of those two sums is an even integer.

- If $(2 \cdot 5^n + k)$ is even, it can be written as $2m$, making $B = 2^n(2m) = m \cdot 2^{n+1}$.
- If $(3 \cdot 5^n + k)$ is even, it can be written as $2m$, making $C = 2^n(2m) = m \cdot 2^{n+1}$.

In either scenario, exactly one of the constructed $(n + 1)$ -digit numbers is divisible by 2^{n+1} . This completes the induction. ■

Q. 2

Problem: For all natural numbers n , define the polynomial $P_n(x) := x^{n+2} - 2x + 1$. (a) Show that the equation $P_n(x) = 0$ has one and only one root c_n in $(0, 1)$. (b) Calculate $\lim_{n \rightarrow \infty} c_n$.

Proof of (a): Notice that $x = 1$ is a trivial root of $P_n(x)$ because $P_n(1) = 1 - 2 + 1 = 0$. We can factor out $(x - 1)$ using the standard geometric sum expansion:

$$P_n(x) = x(x^{n+1} - 1) - (x - 1) = (x - 1)(x^{n+1} + x^n + \cdots + x^2 + x - 1)$$

Let $Q_n(x) = x^{n+1} + x^n + \cdots + x^2 + x - 1$. Any root of $P_n(x)$ in the interval $(0, 1)$ must be a root of $Q_n(x)$, since $x - 1 \neq 0$ on this interval.

Taking the derivative, $Q'_n(x) = (n+1)x^n + nx^{n-1} + \cdots + 1$. For all $x > 0$, $Q'_n(x) > 0$, meaning $Q_n(x)$ is strictly monotonically increasing on $(0, 1)$. Evaluating the boundaries:

$$Q_n(0) = -1 < 0,$$

$$Q_n(1) = n > 0$$

By the Intermediate Value Theorem and strict monotonicity, $Q_n(x)$ has exactly one root $c_n \in (0, 1)$. Consequently, $P_n(x)$ has exactly one root in $(0, 1)$. ■

Proof of (b): From $P_n(c_n) = 0$, we have the relation $2c_n - 1 = c_n^{n+2}$. Since $c_n \in (0, 1)$, it must be that $c_n^{n+2} > 0$, which strictly requires $2c_n - 1 > 0$, implying $c_n > 0.5$.

Next, we determine the sequence's monotonicity. We evaluate $Q_{n+1}(x)$ at c_n :

$$Q_{n+1}(c_n) = c_n^{n+2} + Q_n(c_n) = c_n^{n+2} + 0 > 0$$

Since Q_{n+1} is strictly increasing and $Q_{n+1}(c_{n+1}) = 0$, the fact that $Q_{n+1}(c_n) > 0$ dictates that $c_{n+1} < c_n$. Thus, $\{c_n\}$ is a strictly decreasing sequence bounded below by 0.5, so it must converge to some limit $L \in [0.5, 1)$.

Because $c_n \leq c_1 < 1$ for all n , the term c_n^{n+2} is bounded by c_1^{n+2} , which vanishes as $n \rightarrow \infty$. Taking the limit of both sides of $2c_n - 1 = c_n^{n+2}$ yields:

$$2L - 1 = 0 \implies L = 0.5$$

Thus, $\lim_{n \rightarrow \infty} c_n = 0.5$. ■

Q. 3

Problem: Show that if $a \neq b$, then $\frac{e^b - e^a}{b - a} < \frac{e^a + e^b}{2}$.

Proof: Consider the function $f(x) = e^x$. The second derivative is $f''(x) = e^x$. Since

$e^x > 0$ for all $x \in \mathbb{R}$, the function $f(x)$ is strictly convex everywhere.

For a strictly convex function f , the Hermite-Hadamard inequality states that the average value of the function over an interval $[a, b]$ is strictly less than the average of the function's values at the endpoints of the interval. Without loss of generality, assume $a < b$. By the right-hand side of the Hermite-Hadamard inequality:

$$\frac{1}{b-a} \int_a^b f(x) dx < \frac{f(a) + f(b)}{2}$$

Substituting $f(x) = e^x$ into the integral:

$$\int_a^b e^x dx = \left[e^x \right]_a^b = e^b - e^a$$

Therefore, the average value on the left side is:

$$\frac{1}{b-a} \int_a^b e^x dx = \frac{e^b - e^a}{b-a}$$

Substituting this back into the Hermite-Hadamard inequality directly yields the desired result:

$$\frac{e^b - e^a}{b-a} < \frac{e^a + e^b}{2}$$

By symmetry, this result holds for any $a \neq b$. ■

Q. 4

Problem: Prove that there cannot exist any function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for every $n \geq 2$, $f(f(n-1)) = f(n+1) - f(n)$ holds.

Proof: Assume, on the contrary that such a function exists mapping strictly to the positive integers ($\mathbb{N} = \{1, 2, 3, \dots\}$). Because f maps to $\mathbb{N}$, the term $f(f(n-1)) \geq 1$. Thus, $f(n+1) - f(n) \geq 1$, which proves $f(n)$ is a strictly monotonically increasing function. A strictly increasing positive integer sequence must satisfy $f(k) \geq k$ for all $k \geq 1$.

Applying this to our recurrence:

$$f(n+1) - f(n) = f(f(n-1)) \geq f(n-1)$$

Suppose there exists some integer m such that $f(m) \geq m+2$. Because f is strictly increasing, applying f to both sides preserves the inequality: $f(f(m)) \geq f(m+2)$. Substituting this back into the functional equation for $n = m+1$ gives:

$$f(m+2) - f(m+1) = f(f(m)) \geq f(m+2)$$

This simplifies to $f(m+1) \leq 0$, which violently contradicts the codomain of $\mathbb{N}$.

Therefore, $f(m)$ can never exceed $m+1$. Coupled with the lower bound $f(m) \geq m$, we are severely restricted to $f(m) \in \{m, m+1\}$ for all $m \geq 1$.

Because $f(k)$ can only step up by 1 or 2 with each index increment, $f(n+1) - f(n) \in \{1, 2\}$. By the functional equation, $f(f(n-1))$ is exactly this difference, restricting $f(f(n-1)) \in \{1, 2\}$. However, we already established $f(k) \geq k$. Substituting $k = f(n-1)$, we get $f(f(n-1)) \geq f(n-1) \geq n-1$.

This imposes the boundary $n-1 \leq 2$, implying $n \leq 3$. Since the condition must hold universally for *all* $n \geq 2$, choosing $n = 4$ gives the impossibility $3 \leq 2$. Thus, no such function can exist. ■

Q. 5

Problem: Suppose that a polynomial $P(x) = x^{10} - 10x^9 + 39x^8 + a_7x^7 + \cdots + a_1x + a_0$ has 10 real roots. Prove that all the roots of $P(x)$ lie between -2.5 and 4.5 .

Proof: Let the 10 real roots of the polynomial be $r_1, r_2, \dots, r_{10}$. By Vieta's formulas, we extract the sums of the roots from the highest coefficients:

$$\sum_{i=1}^{10} r_i = 10$$

and,

$$\sum_{1 \leq i < j \leq 10} r_i r_j = 39$$

We calculate the sum of the squares of the roots:

$$\sum_{i=1}^{10} r_i^2 = \left(\sum_{i=1}^{10} r_i \right)^2 - 2 \sum_{i < j} r_i r_j = 10^2 - 2(39) = 100 - 78 = 22$$

To establish the bounds on an arbitrary root r_k , we isolate it. The remaining 9 roots satisfy:

$$\sum_{i \neq k} r_i = 10 - r_k,$$

and

$$\sum_{i \neq k} r_i^2 = 22 - r_k^2$$

We apply the Cauchy-Schwarz inequality to the 9-dimensional vectors $\vec{u} = (1, 1, \dots, 1)$

and $\vec{v} = (r_1, \dots, r_{k-1}, r_{k+1}, \dots, r_{10})$:

$$\left(\sum_{i \neq k} 1 \cdot r_i \right)^2 \leq \left(\sum_{i \neq k} 1^2 \right) \left(\sum_{i \neq k} r_i^2 \right)$$

$$\Rightarrow (10 - r_k)^2 \leq 9(22 - r_k^2)$$

Expanding and combining like terms:

$$100 - 20r_k + r_k^2 \leq 198 - 9r_k^2$$

$$\Rightarrow 10r_k^2 - 20r_k - 98 \leq 0 \implies 5r_k^2 - 10r_k - 49 \leq 0$$

Using the quadratic formula, the boundary roots of $5x^2 - 10x - 49 = 0$ are:

$$x = \frac{10 \pm \sqrt{100 - 4(5)(-49)}}{10} = \frac{10 \pm \sqrt{1080}}{10} = 1 \pm \sqrt{10.8}$$

Since $3 < \sqrt{10.8} < 3.5$, the roots r_k are tightly constrained within the interval $[1 - 3.5, 1 + 3.5] = [-2.5, 4.5]$. This applies symmetrically to any root, proving all 10 roots lie between -2.5 and 4.5 . ■

Q. 6

Problem: Let a sequence of real numbers $\{a_n\}_{n \geq 1}$ be defined as $a_n := \sum_{k=1}^n \frac{1}{k^3}$. Prove that $\sum_{k=1}^n \frac{1}{k^3 a_k^2} < \frac{6}{5}$.

Proof: We analyze the sequence a_n . Since $a_1 = 1$, the first term of our given summation is strictly $1/(1^3 \cdot 1^2) = 1$. For all $k \geq 2$, a_k is strictly increasing. Thus, $a_k \geq a_2 = 1 + \frac{1}{8} = \frac{9}{8}$. Squaring this implies $a_k^2 \geq \frac{81}{64}$, meaning $\frac{1}{a_k^2} \leq \frac{64}{81}$ for all $k \geq 2$.

We can bound the summation tail by factoring out this maximum coefficient:

$$\sum_{k=2}^n \frac{1}{k^3 a_k^2} \leq \frac{64}{81} \sum_{k=2}^n \frac{1}{k^3}$$

To cleanly evaluate $\sum_{k=2}^n \frac{1}{k^3}$, we bind it with a telescoping fraction by observing that $k^3 > k^3 - k = (k-1)k(k+1)$:

$$\frac{1}{k^3} < \frac{1}{(k-1)k(k+1)} = \frac{1}{2} \left(\frac{1}{k-1} - \frac{2}{k} + \frac{1}{k+1} \right) = \frac{1}{2} \left[\left(\frac{1}{k-1} - \frac{1}{k} \right) - \left(\frac{1}{k} - \frac{1}{k+1} \right) \right]$$

Summing this from $k = 2$ to n collapses everything except the outer boundaries:

$$\sum_{k=2}^n \frac{1}{k^3} < \frac{1}{2} \left[\left(\frac{1}{1} - \frac{1}{2} \right) - \left(\frac{1}{n} - \frac{1}{n+1} \right) \right] < \frac{1}{2} \left(\frac{1}{2} \right) = \frac{1}{4}$$

Plugging this integral upper bound back into our inequality gives:

$$\sum_{k=2}^n \frac{1}{k^3 a_k^2} < \frac{64}{81} \times \frac{1}{4} = \frac{16}{81}$$

Adding the $k = 1$ term back in, the total sum is bounded by $1 + \frac{16}{81} = \frac{97}{81}$. Since $97 \times 5 = 485$ and $81 \times 6 = 486$, it holds that $\frac{97}{81} < \frac{6}{5}$. ■

Q. 7

Problem: Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuously differentiable function such that $\int_0^1 f(x) dx = 1$. Prove that $\int_0^1 (1 + x^2) f^2(x) dx \geq \frac{4}{\pi}$.

Proof: We use the integral form of the Cauchy-Schwarz inequality. First we multiply the integrand of our constraint by 1 in the form of $\frac{\sqrt{1+x^2}}{\sqrt{1+x^2}}$:

$$\int_0^1 f(x) dx = \int_0^1 \left(f(x) \sqrt{1+x^2} \right) \left(\frac{1}{\sqrt{1+x^2}} \right) dx$$

Applying the Cauchy-Schwarz inequality for integrals to these two newly grouped functions:

$$\left(\int_0^1 f(x) dx \right)^2 \leq \left(\int_0^1 \left(f(x) \sqrt{1+x^2} \right)^2 dx \right) \left(\int_0^1 \left(\frac{1}{\sqrt{1+x^2}} \right)^2 dx \right)$$

Squaring the internal terms and substituting our initial condition $\int_0^1 f(x) dx = 1$:

$$1^2 \leq \left(\int_0^1 (1+x^2) f^2(x) dx \right) \left(\int_0^1 \frac{1}{1+x^2} dx \right)$$

The second integral is a standard trigonometric integration:

$$\int_0^1 \frac{1}{1+x^2} dx = \left[\arctan(x) \right]_0^1 = \frac{\pi}{4} - 0 = \frac{\pi}{4}$$

Substitute this back to finalize the proof:

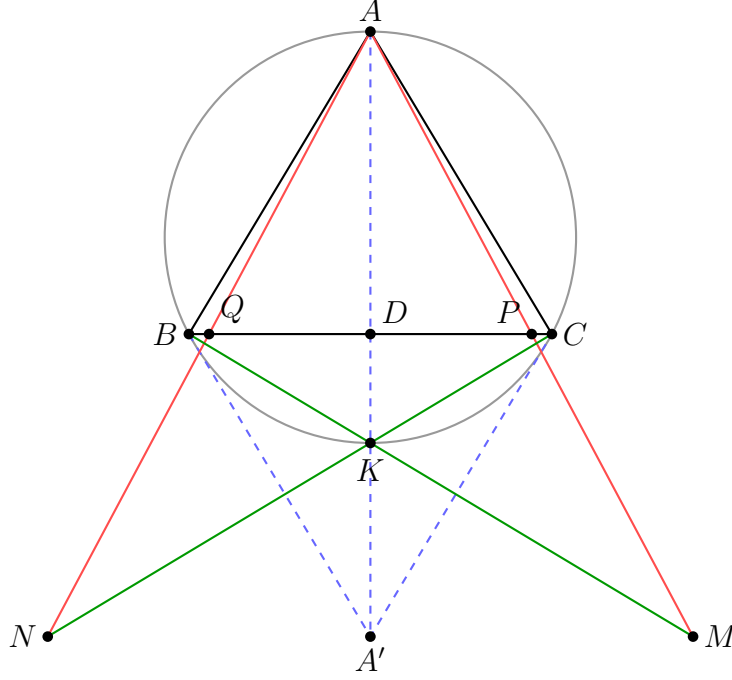
$$1 \leq \left(\int_0^1 (1+x^2) f^2(x) dx \right) \left(\frac{\pi}{4} \right) \implies \int_0^1 (1+x^2) f^2(x) dx \geq \frac{4}{\pi}$$

■

Q. 8

Problem: Suppose that points P and Q lie on side BC of acute-angled $\triangle ABC$ such that $\angle PAB = \angle ACB$ and $\angle CAQ = \angle ABC$. Let M and N be the points on lines AP and AQ , respectively, such that P is the midpoint of AM , and Q is the midpoint of AN . Prove that the intersection of BM and CN lies on the circumcircle of $\triangle ABC$.

Proof:



Let the sides of $\triangle ABC$ be a, b, c and its angles be A, B, C . Consider $\triangle PAB$ and $\triangle CAB$. They share the angle $\angle B$, and by definition, $\angle PAB = \angle C$. Thus, $\triangle PAB \sim \triangle CAB$. This similarity yields the ratios: $\frac{PA}{CA} = \frac{AB}{CB} \implies PA = \frac{bc}{a}$, and $\frac{PB}{AB} = \frac{AB}{CB} \implies PB = \frac{c^2}{a}$. Furthermore, it implies $\angle APB = \angle BAC = A$. Since P lies strictly between A and M on a line, the supplementary angle $\angle BPM = 180^\circ - A$. Because P is the midpoint of AM , $PM = PA = \frac{bc}{a}$.

Let D be the midpoint of side BC . Extend the median AD to point A' such that $AD = DA'$. This constructs a parallelogram $ABA'C$, meaning $BA' = b$ and $CA' = c$. In $\triangle ABA'$, the sides are $AB = c$, $BA' = b$, and the included angle is $\angle ABA' = 180^\circ - A$ (as adjacent angles in a parallelogram are supplementary). Now, compare $\triangle BPM$ and $\triangle ABA'$:

- Their included angles are identical: $\angle BPM = \angle ABA' = 180^\circ - A$.
- The ratio of the sides around this angle in $\triangle BPM$ is $\frac{PB}{PM} = \frac{c^2/a}{bc/a} = \frac{c}{b}$.
- The ratio of the corresponding sides in $\triangle ABA'$ is $\frac{AB}{BA'} = \frac{c}{b}$.

Because the ratios are equal, $\triangle BPM \sim \triangle ABA'$ by SAS similarity. This tells us that $\angle MBP = \angle A'AB$. Since A' lies on the extension of the median AD , this is equivalent to $\angle MBP = \angle DAB$.

By entirely symmetric reasoning involving Q and N , we can establish that $\triangle CQN \sim \triangle ACA'$. This gives $\angle NCQ = \angle A'AC = \angle DAC$.

Let K be the intersection of BM and CN . In $\triangle KBC$, we compute the angle at K :

$$\angle BKC = 180^\circ - (\angle KBC + \angle KCB)$$

Using the equivalencies derived above, we substitute:

$$\angle BKC = 180^\circ - (\angle DAB + \angle DAC) = 180^\circ - \angle BAC = 180^\circ - A$$

Because $\angle BKC$ and $\angle BAC$ sum to 180° , and points A and K lie on opposite sides of the segment BC (since extending past P and Q places M and N outside the original triangle), the quadrilateral $ABKC$ is cyclic. Thus, point K lies on the circumcircle of $\triangle ABC$. ■
